

VU Medical Zone

BT 101 Final Term Subjective

Q1. Define Survivorship curve. 2 Marks

Survivorship curve is one tool. It measures the proportion of individuals in a given species that are alive at different ages.

Q2. When Adaptation occurs? 2 Marks

Adaptation occurs when a change in a phenotype increases an animal's chance of successful reproduction.

Change in environment is the force behind adaptations.

It may result in the evolution of multiple new groups.

Q3. Function of Amnion in amniotic egg? 2 Marks

First vertebrates to possess amniotic eggs. Amnion protects embryo from desiccation, cushion the embryo, promote gas transfer, and store waste materials.

Q4. Factors that affect Logistic growth. 3 Marks

Environmental factors affect population growth.

- Climate
- Food
- Space
- Competition

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Q5. Explain Fitness with examples. 3 Marks

Adaptedness or fitness: is a measure of the capacity for successful reproduction in a given environment. Not every characteristic is an adaptation to some kind of environmental situation. Neither adaptation leads to perfection. Genes of all traits are present in a population in form of alleles.

Examples:

Two phenotypes of toads

1. Green
 2. Brown • Green leave average 4 offspring:
 3. Brown leave average 2.5 offsprings.
- By custom, the most fit phenotype is assigned a fitness value of 1.0.

Q6. Characters of Nematoda? 5 Marks

- Triploblastic • Bilateral • Vermiform • Unsegmented • Pseudocoelomate • Complete digestive tract • Unique excretory system comprised of two or more renette cell • Body wall has only longitudinal muscles • Cuticle may contain spines, bristle papillae and ridges • Pseudocoelom is a spacious fluid filled cavity • Visceral organs • Form hydrostatic skeleton • May be herbivores, carnivores, omnivores and saprotrophs • Feed on blood and tissue fluid of their hosts • Complete digestive tract • Excretion of nitrogenous waste • Ventral gland cells, called renette • Nervous system consist of anterior nerve ring.

Q7. Process of Decomposition. 5 Marks

Decomposition

1. Autolysis Breaking down of tissues by the body's own internal chemicals and enzymes
2. Putrefaction • Breakdown of tissues by bacteria. • release compounds such as cadaverine and putrescine causing decaying odor • Proteolysis releases nitrogen and phosphorous • Carbohydrate breakdown -

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Aerobic by fungi and bacteria - Anaerobic by bacteria • Anerobic degradation of lipids • Nucleic acids and bones degradation.

Q8. Effects of Global Warming. 5 Marks

Arctic sea ice shrinking • Decrease in extent – 2.7% per decade since 1978 • Max. area of seasonally frozen ground decreased by 7% in Northern Hemisphere since 1900.

- Global sea level rise • Average rate of 1.8 mm per year (1961-2003) • Likely Sea level rise during the 21st century - 5 mm per year • Low lying areas, small island states may disappear, salt water intrusion into aquifers, floods • Low-lying coastal cities will be at the forefront of impacts

Non-polar glaciers • Widespread retreat during the 20th century

- Continental Precipitation • Significant increase in North & South America, northern Europe, north & central Asia • Drying in Mediterranean, southern Africa, parts of South Asia

- Extreme Weather • Frequency of drought, heat waves, typhoons, hurricanes and cyclones increased from 1960 until 2010. • Extremely wet or dry events within the monsoon period have increased since 1980.

Q9. Structure and Function of Golgi Apparatus. 5 Marks

Structure

• Located near the end of the ER close to the nucleus • Composed of several layers of cisternae (fluid-filled sacs) • Has three different parts: 1. cis-Golgi 2. medial-Golgi 3. trans-Golgi

Functions

• cis is first cisternae. • trans is final cisternae. • From trans, proteins are packaged into vesicles destined to lysosomes, secretory vesicles, or the cell surface. Transport vesicles are used to move back and forth between the ER and Golgi bodies • One side of the Golgi body receives transport vesicles produced by the ER. • This side is called cis face. • Opposite side releasing vesicle is trans face.

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Q10. Composition of Atmosphere?

10 Marks

Comprised of a mixture of:

1. Invisible permanent gases
 2. Variable gases
 3. Suspended microscopic particles
1. Permanent Gases Form a constant proportion of the total atmospheric mass
 2. Variable Gases Distribution and concentration varies in space and time
 3. Aerosols Suspended particles and liquid droplets (excluding cloud droplets)

1. Permanent Gases

78% Nitrogen (N₂) 21% Oxygen (O₂) Relative percentages of the permanent gases remain constant up to 80 - 100km high. Turbulent mixing causes atmospheric composition to be fairly homogenous. • Called as Homosphere

2. Variable Gases

• Water vapor (H₂O): 0 to 4% • Carbon Dioxide (CO₂): 0.038% • Methane (CH₄): 0.00017% • Ozone (O₃): 0.000004%

Variable Water vapors

• Water vapor is invisible. • Its not cloud droplets. • 3rd major component • Surface percentages vary between

3. Aerosols

• Small (or “tiny”) solid particles or liquid droplets (excluding clouds and rain) • Aerosols can be manmade (anthropogenic) or naturally occurring (like ocean salt, dust, plant emissions).

Q11. Define Mesosphere.

2 Marks

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It is a Layer, directly above the stratosphere and directly below the thermosphere • Lower boundary is 50 to 65 kilometres • Upper boundary is around 85 to 100 kilometres • It is called as “near space”.

Q12. Define Thylakoid. 2 Marks

Photosynthesizing bacteria that are related to the chloroplasts of eukaryotic plants and algae. Also known as blue green algae. Produce energy by photosynthesis. They have flattened sacs called thylakoids where photosynthesis is performed.

Q13. Peroxisome. 3 Marks

Isolated from liver cells by centrifugation – Called them peroxisomes because they generate and destroy H_2O_2 . Peroxisome, membrane bound organelle occurring in the cytoplasm of eukaryotic cells.

Functions • Contains oxidative enzymes • Break down fatty acids and amino acids by oxidation. • Long fatty acids are converted to medium chain fatty acids.

Q14. Countershading. 3 Marks

Shadow makes upper side darker and underside lighter. Countershading graded color to counteract the effect of self-shadowing. Creating an illusion of flatness. Makes animal darker below and lighter above
Example: gazelles, grasshoppers, sharks, dolphins.

Q15. Evolutionary force and Gene flow. 5 Marks

The amount of genetic variation in a population may be determined by the relative strength of different evolutionary processes. Sometimes these processes act together, and in other cases they work in opposition.

Gene flow may promote or constrain evolutionary change:

- Gene flow can be either a constructive or a constraining force.

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- It can spread a beneficial mutation that arises in one population to other populations.
- Gene flow can impede adaptation within a population by the continual flow of inferior alleles from other populations.
- Natural selection is removing inferior alleles while gene flow can bring them back.

Example: Heavy metals tolerance

Q16. Water waste treatment.

5 Marks

Septic Tanks

- Bacteria in sewage degrade organic matter
- Tank buried in ground to treat sewage from an individual home
- Wastewater flows into tank
- Solid material (“sludge”) settles to bottom of tank
- Clean water flows out of tank into ground through subsurface drains.

Lagoons (Stabilization Ponds) • 3-5’ deep • Sunlight, algae, oxygen interact to clean wastewater.

Conventional Sewage Treatment • Processes 1. Collection 2. Treatment

Conventional Sewage Treatment Collection • Combined Sewers Carry stormwater and wastewater •

Separated Sewers 1. Sanitary Sewers 2. Storm Sewers.

Primary Treatment • Mechanical process • Screening • Grinding • Grit Chamber (heavy particles - “grit”) • Sedimentation Tank (suspended solids - “sludge”) • Chlorination of effluent

Secondary Treatment • Biological process • Uses bacteria to remove organic matter dissolved in water (ODW) and lower BOD Types • Trickling Filter • Activated Sludge.

Q17. Exponential Growth?

10 Marks

Exponential Growth • Resources are abundant. • Remarkable potential to increase in number • Exponential addition in population size • Population may grow exponentially to certain time • Ultimately limited by resource availability • Formula of exponential growth $dN = rN$ $N =$ Population size dt $r =$ intrinsic rate of natural increase.

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Factors affecting reproductive rate • Number of offspring produced • Likelihood of survival to reproductive age • Duration of the reproductive period • Length of time to reach maturity • Environmental factors

Q18. Factors Affecting the reproductive rate? 2 Marks

Factors affecting reproductive rate • Number of offspring produced • Likelihood of survival to reproductive age • Duration of the reproductive period • Length of time to reach maturity • Environmental factors.

19. Acid Rain. 5 Marks

Form of precipitation that is unusually acidic. • It has elevated levels of hydrogen ions (low pH). • Deposition of a mixture

1. Wet (rain, snow, sleet, fog, cloud water and dew)
2. Dry (acidifying particles, dust and gases) acidic components.

• Unpolluted rain has an acidic pH, but usually no lower than 5.7. • This low pH is due to carbonic acid $\text{H}_2\text{O} + \text{CO}_2 \rightleftharpoons \text{H}_2\text{CO}_3 (\text{aq})$ • Nitric acid is produced as result of lightening.

Causes • Natural • Volcanic eruption releases SO_2 in air. • Nitrogen fixation adds NO_2 in air. • Electrical activity in the atmosphere such as lightning produces NO_2 • Acid deposits in glacial ice.

• Human Activities • Sulfur and nitrogen compounds from human sources • Electricity generation, factories, and motor vehicles • Greatest contribution by burning coal in power plants.

Effects • Not many things can grow in acidic conditions • Low pH and high aluminum concentrations can damage or kill fish and aquatic populations • At pH lower than 5 most fish eggs will not hatch and lower pH can kill adult fish.

Q20. Effects of Acid Rain? 10 Marks

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Effects • Not many things can grow in acidic conditions • Low pH and high aluminum concentrations can damage or kill fish and aquatic populations • At pH lower than 5 most fish eggs will not hatch and lower pH can kill adult fish. • Soils can be damaged by the hydronium ion, which mobilizes aluminum and encourages leaching of minerals such as magnesium essential for plant life. • Forests suffer from soil damage however most food crops are unharmed because the nutrients lost are replaced in fertilizer. • Monuments made of Calcium Carbonate (limestone and marble) will react with acid rain to form Gypsum • Increases the oxidation rate of metals such as copper and bronze.

Q21. Feeding Level? 5 Marks

Feeding Levels • Food energy passes through different sequence. • Example: plants → grass hopper → frog → snake → hawk • Many plant species eaten by insects • Many insects species eaten by frogs.

Trophic Level • Trophic level of an organism is position it occupies in a food chain. Food Chain • Linear network of links to pass energy in form of food • Simplified abstractions of real food webs, but complex in their dynamics. Feeding Levels • This complexity is organized by ecologists. • Recognized limited number of feeding or trophic levels.

Q22. Difference between oscillating and Frequency dependent selection. 5 Marks

Oscillating Selection • In some cases, selection favors one phenotype at one time and another phenotype at another time. • Example: Medium ground finches of the Galápagos Islands • In times of drought, the supply of small, soft seeds is depleted. • But there are still enough large seeds around. • Consequently, birds with big bills are favored. • However, when wet conditions return, the ensuing abundance of small seeds favors birds with smaller bills.

Difference between oscillating and Frequency dependent selection

• In oscillating selection, the fitness of a phenotype does not depend on its frequency. • Rather, environmental changes lead to the oscillation in selection. • In frequency dependent selection, it is the change in frequencies themselves. • Fitness of the different phenotypes can vary according to their frequencies and environment.

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Q23. Conventional aviation? 2 Marks

• The lowest layer • All weather conditions take place here. • Conventional aviation takes place here. • It contains: • 75% of the atmosphere's mass • 99% of the total mass of water vapor and aerosols.

Q24. Types of pseudopodia? 5 Marks

Types of pseudopodia

• Majorly amoeba has four types

1. Lobopodia

2. Filopodia

3. Reticulopodia

4. Axopodia

1. Lobopodia • Contain ectoplasm and endoplasm • Used for locomotion • Used for engulfing feed.

2. Filopodia • Only ectoplasm in pseudopodia • Two-way streaming • Provides food in conveyor belt fashion.

3. Reticulopodia • Similar to filopodia • Branch and rejoin to form net like series of cell extensions.

4. Axopodia • These are thin filamentous supported by central axis of microtubules. • Food caught on axopodia delivered to the central cytoplasm of amoeba.

Q25. Ciliophora reproduction. 5 Marks

Reproduction

Asexually reproduction is common. • Cell divided into two by mitosis through: • Binary fission • Budding.

Sexual reproduction

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- Occurs when placed under stress
- Paramecia may engage in a process known as conjugation
- Exchange genetic material with other individuals
- Process may take many hours.

Q26. Prevention of Animal from Dehydration. 2 Marks

Adaptations To avoid dehydration:

Q27. Aneuploidy and Polyploidy. 3 Marks

1. Aneuploidy • Nondisjunction at a single chromosome result in an abnormal number of chromosomes.
2. Polyploidy • Product of nondisjunction during meiosis which results in additional copies of the entire genome.

Q28. Composition of Matrix in Mitochondria? 3 Marks

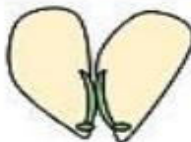
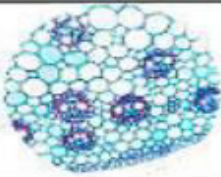
Matrix

- Enclosed in inner membrane
- Gel like consistency, Dense, homogenous.
- Contains 2/3rd of total protein of mitochondria.
- Contains enzymes, DNA genome, ribosomes, tRNA, granules, fibrils and tubules.

Q29. Monocot and Dicot? 5 Marks

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MONOCOT		DICOT	
Single Cotyledon		Two Cotyledon	
Long Narrow Leaf Parallel Veins		Broad Leaf Network of Veins	
Vascular Bundles Scattered		Vascular Bundles in a Ring	
Floral Parts in Multiples of 3		Floral Parts in Multiples of 4 or 5	

Q30. Adaptedness. 3 marks

- Adaptedness or fitness: is a measure of the capacity for successful reproduction in a given environment.
- Not every characteristic is an adaptation to some kind of environmental situation.
- Neither adaptation leads to perfection.

Q31. Ways of Reproductive isolation. 10 Marks

Ways of Reproductive Isolation

1. Premating isolation

- Geographical Isolation - prevents mating from taking place
- Barriers, such as rivers or mountain ranges, may separate subpopulations.

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- Mechanical Isolation - If courtship behavior patterns of two animals are not mutually appropriate for each other.
- Mating does not occur.
- Temporal Isolation – type of reproductive isolation due to reproducing at different times.
- Behavioral Isolation – reproductive isolation due to differences in mating or courtship behavior.
- Even though they are able to physically interbreed.
- Example: Different ways to decorate nests to attract females.
- Satin bowerbird & MacGregor's Bowerbird.
- Ecological Isolation – reproductive isolation due to living in different parts of the habitat (microhabitats).
- Rarely come in contact with each other.
- Two closely-related birds of the Turdus genus.
- Blackbird (Turdus merula).
- Ring ouzel (Turdus torquatus).

2. Postmating isolation

- Prevents successful fertilization and development.
- Even though mating may have occurred. Example:
- Conditions in the reproductive tract of a female may not support the sperm of male.
- Though successful fertilization Example:
- Mule produced from a mating of a male donkey and a mare is a sterile hybrid.

Q32. Bottleneck Effect? 2 Marks

Bottleneck Effect

- It occurs when number of individuals in a population is drastically reduced.
- Mostly by any disaster e.g. natural catastrophe, over hunting, habitat damage.
- Survivors have gene pool not representing the initial population.
- Decreases in genetic diversity make populations less likely to withstand environmental stress.
- More susceptible to extinction.

Q33. Balanced polymorphism? 5 Marks

Balanced Polymorphism

- It occurs when different phenotypes are maintained at relatively stable frequencies in the population.
- Resemble a population in which disruptive selection operates. Example:
- Sickle cell anemia is homozygous recessive trait.
- Most common in Africa where malaria is common as well.
- Heterozygous

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alleles carriers are less more susceptible to malaria. • Homozygous dominant alleles carriers are more susceptible to malaria. • Homozygous recessive alleles carriers have low survival due to anemia. • Therefore, heterozygous are more successful in survival and reproductive potential than two others. • Heterozygote superiority. • It leads to balanced polymorphism • Because perpetuation of the alleles in the heterozygous condition maintains both alleles at a higher frequency • Frequency of homozygous phenotypes will be lower than heterozygous. • This is Balanced polymorphism.